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INSOMNet: Automated insomnia detection using scalogram and deep neural networks with ECG signals

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ABSTRACT

Sleep is a natural state of rest for the body and mind. It is essential for a human's physical and mental health because it helps the body restore itself. Insomnia is a sleep disorder that causes difficulty falling asleep or staying asleep and can lead to several health problems. Conventional sleep monitoring and insomnia detection systems are expensive, laborious, and time-consuming. This is the first study that integrates an electrocardiogram (ECG) scalogram with a convolutional neural network (CNN) to develop a model for the accurate measurement of the quality of sleep in identifying insomnia. Continuous wavelet transform has been employed to convert 1-D time-domain ECG signals into 2-D scalograms. Obtained scalograms are fed to AlexNet, MobileNetV2, VGG16, and newly developed CNN for automated detection of insomnia. The proposed INSOMNet system is validated on the cyclic alternating pattern (CAP) and sleep disorder research center (SDRC) datasets. Six performance measures, accuracy (ACC), false omission rate (FOR), sensitivity (SEN), false discovery rate (FDR), specificity (SPE), and threat score (TS), have been calculated to evaluate the developed model. Our developed system attained the classifications ACC of 98.91%, 98.68%, FOR of 1.5, 0.66, SEN of 98.94%, 99.31%, FDR of 0.80, 2.00, SPE of 98.87%, 98.08%, and TS 0.98, 0.97 on CAP and SDRC datasets, respectively. The developed model is less complex and more accurate than transfer-learning networks. The prototype is ready to be tested with a huge dataset from diverse centers.

1. Introduction

Sleep is a physiological process essential for the proper functioning of the body and mind. It is characterized by a decreased heart rate, lower body temperature, and a decrease in the activity of the brain and other organs [1]. Sleep is essential for various reasons, including physical and mental strength, cognitive function, and overall well-being. Adequate sleep helps to repair and rejuvenate the body, allowing the immune system to function appropriately, promoting growth and development, and helping to regulate hormones. It also helps to improve mood, and cognitive function, including memory and concentration [1]. Adults require roughly 7-9 hours of sleep each day for healthy lives. The current figures reveal that nearly 30% of Americans sleep six hours or less, 33% of adults have sleep difficulties in India, and approximately 10% of the world's population suffers from various sleep-related disorders [2,3]. Insomnia is a common sleep problem characterized by

trouble in sleeping or staying asleep or both [4]. It can lead to many negative consequences, including fatigue, difficulty in concentrating, emotional disruptions, and impaired ability to perform daily activities, which may cause a variety of cognitive and medical problems, including depression and anxiety disorders, [4]. Insomnia can be acute and may lead to underlying medical or psychological conditions [4]. Health professionals analyze the severity of insomnia by observing the sleeping patterns of patients using questionnaires and sleep diaries. It tracks an individual's sleep patterns and habits over a while. It can be used to track the amount of sleeping time, the time it takes to fall asleep, and the number of awakenings during the night. Doctors sometimes suggest a sleep study of a patient. Sleep latency test (SLT) and polysomnography (PSG) recordings are the most common way to examine the sleep study. In SLT, a health worker measures the amount of time taken by an individual to fall asleep [5]. The PSG is a non-invasive study that records various biometric data during sleep, including eye

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movements, brain signals, heart rate, muscle activity, and breathing patterns. PSG measurements are used when the initial examination is insufficient due to the existence of secondary sleep disorders, either pharmaceutical or behavioral. During both SLT and PSG tests, a patient must spend a few nights in a sleep laboratory. However, the measuring setup for PSG recordings is time-consuming and uncomfortable for the patient because numerous wires and electrodes are attached to the body [6]. Insomnia detection using existing techniques is an inconvenient and expensive process. Therefore, an affordable, simple, less complex, and effective solution is desirable. Research has found that insomnia patients are more likely to have higher sympathetic nervous system activity levels, shown in the electrocardiogram (ECG) patterns as a change in the increased QT and R-R intervals [7]. The ECG acquisition is a simpler, potentially cost-effective method for detecting insomnia when compared to OSG or EEG-based systems. ECG signals were found to be useful in the monitoring of sleep disorders by Bsoul et al. [8]. Hence, ECG can be an alternative solution to diagnose insomnia accurately at a low cost. Manual screening of insomnia through ECG data is time-consuming, labor-intensive, and subject to human mistakes. Consequently, there is an urgent requirement for an intelligent system that can effectively identify the changes in ECG patterns due to insomnia. In [9], we developed a wavelet-based system for the automated detection of CAP phases and sub-phases using EEG signals. Now, we are extending this method to insomnia detection. In this article, we present a novel, accurate, and robust automated system for the precise detection of insomnia via integrating ECG scalograms with a convolutional neural network (CNN) approach. Continuous wavelet transform (CWT) has been employed to obtain scalograms from 1-D ECG signals. Obtained scalogram images are fed as an input to various benchmarks as well as newly developed lightweight, less complex CNN for the accurate classification of ECG scalograms into normal and insomnia groups. The following are the main contributions of the proposed approach:

- This is the first work to present an intelligent system that integrates an ECG scalogram with a CNN for the automated identification of insomnia.
- Our proposed INSOMNet architecture has yielded high performance in detecting insomnia using two sleep datasets.
- The process of selecting and categorizing the most appropriate features is time-consuming and subjective. A less complex, deep CNN coupled with T-F scalograms of ECG signals has been employed to address this.
- Presented network attained higher performance compared to benchmarked models and all currently used cutting-edge approaches.
- Our developed prototype is suitable to design wearables for insomnia screening and improve the quality of life.

The remainder of this article is arranged as follows: various insomnia detection strategies, the cyclic alternating pattern (CAP), sleep disorder research center (SDRC) datasets, and the methodologies used are given in Section 2. The experimental findings are shown in Section 3. The results of this work are discussed in Section 4. The summary of the work and potential future directions are depicted in Section 5.

2. Materials and methods

Numerous approaches for detecting insomnia using electroencephalogram (EEG), electrooculography (EoG), ECG, magnetic resonance imaging (MRI), piezoelectric, pressure sensors, etc., have been investigated in the literature. Xu et al. [10] presented a review study on the automated diagnosis of sleep problems. They examined the application of various machine and deep learning methods for the screening of sleep disorders using different physiological signals. Bahrami et al. [11] presented a comparative analysis of machine and deep-learning algorithms for the detection of sleep disorders using ECG signals. Excessive

use of cell phones and the internet are some of the factors responsible for sleep disorders [12]. In [13], authors examined the psychometric measures of the insomnia severity index (ISI) to identify insomnia cases in a population-based sample. Farina et al. [14] examined heart rate variability (HRV) signals to diagnose insomnia patients. The decreased HRV was seen in insomnia due to an imbalance in the autonomic nervous system, with increased activity of the sympathetic and decreased activity of the parasympathetic nervous system. A deep learning method for insomnia detection using 57 EEG features has been presented by Shahin et al. [15]. They utilized only 2 EEG channels for precise detection of insomnia subjects and reported an accuracy of 92%. An optimal bi-orthogonal wavelet (BW) transform approach using single-channel EEG data has been presented by Sharma et al. [15]. Their developed model yielded an accuracy of 95.6%. In [2], authors proposed an optimal antisymmetric BW filter bank for efficient decomposition of ECG signals into wavelet subbands and computed $L - 1$ norm features from each subband. They employed a K-nearest neighbor (KNN) classifier for the identification of insomnia patients and reported an accuracy of 97.87%. Smoothed Gabor spectrogram of ECG data and deep learning-based sleep disorder detection strategy has been presented by Gupta et al. [16]. Their developed model yielded an accuracy of 94.81%. An unsupervised feature learning algorithm with ECG data has been proposed to detect sleep apnea by Feng et al. [17]. Their developed system yielded a detection accuracy of 85.1%. Angelova et al. [18] extracted sample entropy and Poincare plot from an actigraphy time series. They employed support vector machine (SVM) and random forest (RF) classifiers to identify insomnia patients. Their developed model attained an accuracy of 84% using RF and 73% using the SVM classifier. Yang et al. [19] investigated the impact of functional connectivity (FC) features and amplitude of low-frequency fluctuation (ALFF) on the identification of chronic insomnia utilizing resting-state functional MRI and logistic regression approach. Their obtained results indicated that FC and ALFF features can differentiate the normal and insomnia patients, and the developed model attained a discrimination accuracy of 86.40%. In [20], authors explored two features-based models to differentiate between insomnia and normal subjects using SVM classifier. Their first model utilized 57 features obtained from 2 channels (C3 and C4) of EEG signals and reported an accuracy of 81%. Their second model employed 15 features evaluated from a hypnogram and yielded an accuracy of 74%. Singular spectrum analysis has been investigated to detect the oscillatory variations in the EEG during sleep [21]. The authors concluded that the EEG test may support the scientific evidence for mental illness. According to research by Kaplan et al. [22], the A1-A2 channels are utilized in the automated classification of sleep-wake stages.

These reported insomnia detection techniques have a variety of shortcomings which results in performance degradation. Utilization of EEG signals for screening sleep disorders requires a specific channel selection approach, which is also a tedious task. A range of machine learning algorithms with varying parameters has been employed in previous work. The choice of appropriate algorithms is another difficult task. The available techniques have reported not high classification accuracy, and also very few studies are available in the literature to detect insomnia using ECG signals. Hence, we have proposed an ECG scalogram-CNN intelligent system for the precise screening of insomnia patients. In the presented insomnia screening approach, ECG1-ECG-2 data are extracted from the publicly available PSG recordings of SDRC and CAP datasets and segmented into fragments of 256 and 512 samples. CWT is employed to transform the ECG segments into scalogram images. Obtained ECG scalograms are fed to deep neural networks. In this work, Alex-Net, MobileNetV2, VGG16, and our developed CNN are utilized for the accurate classification of ECG scalograms into normal and insomnia subjects. Fig. 1 represents the schematic diagram of the proposed INSOMNet methodology.

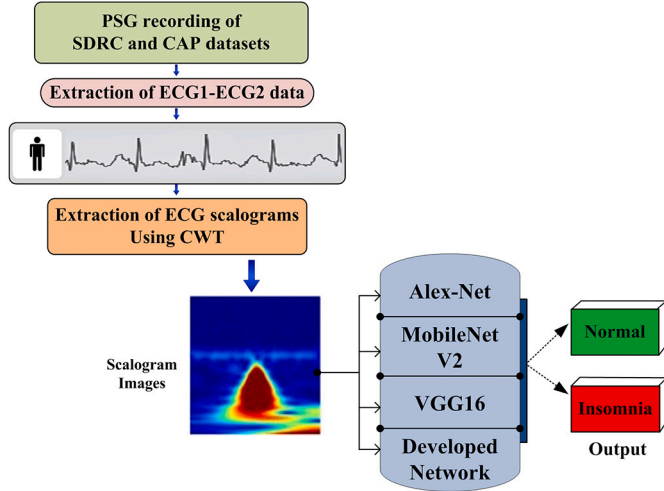


Fig. 1. Proposed INSOMNet methodology.

2.1. Dataset

Publicly accessible CAP [23] and SDRG [24] sleep datasets are used to validate the presented approach. The CAP sleep dataset is a set of 108 PSG recordings, including 92 recordings from patients with sleep disorders such as narcolepsy, insomnia, and bruxism, and 16 recordings from healthy subjects. The dataset is available in “.edf” file format and has been analyzed using the CAP method. The CAP classification system is used to classify different stages of sleep, understand sleep disorders, and improve sleep quality. The records in the dataset include information such as the duration and frequency of each stage of sleep, as well as other characteristics such as the presence of sleep-disordered breathing or movements during sleep, 2 channels electrooculogram (EOG), 3 channels electroencephalogram (EEG) (F3 or F4, C3 or C4 and O1 or O2, referred to A1 or A2), 2 channels ECG, electromyogram (EMG), and other respiration data (abdominal and thoracic effort, airflow signs, and SpO_2). The data were recorded using different ECG channels at different sampling frequencies. In this study, we have utilized only ECG1-ECG2 channels with a sampling frequency of 512 Hz.

The SDRG sleep dataset comprises the recording of 22 subjects including 14 females and 8 males. The age of participants was in the range of 18 to 63 years. The dataset was composed of 11 healthy subjects and 11 psychophysiological insomnia patients. The data were collected at a sampling ratio of 256 Hz from 14 EEG (C4A1, C3A2, F3, F4, C3, C4, A1, A2, O1, O2, F3A2, F4A1, O1A2, O2A1), 3 EMG (MG, EMG1, EMG2), 6 EOG (EOG1, EOG2, EOG1A1, EOG2A1, EOG1A2, EOG2A), and ECG channel. For data acquisition, all participants underwent a whole night PSG test using the SOMNOscreen plus PSG device manufactured by SOMNOmedics GmbH, Germany. The test lasted 8 hours (23:00-07:00 h), as per the established standard at the SDRG of KUMS in Iran. This work utilized only ECG channel data with a sampling frequency of 256 Hz. A standard 16-bit analog-to-digital converter was used in both CAP and SDRG data to digitize the signals.

Fig. 2 and 3 depict the typical ECG signals of two classes (a: normal; b: insomnia) obtained from CAP and SDRG dataset, respectively.

2.2. Continuous Wavelet Transform (CWT)

Biological time sequences are non-stationary and nonlinear in nature. Extraction of in-depth details from a non-stationary signal is a time-consuming and difficult process. Time-frequency representation (TFR) of a biological time sequence contains more insight information about the signal [25]. These images show the variation of amplitude and frequency with time. This study employed CWT to transform 1-D ECG signals into its TFR. The CWT is a mathematical technique used

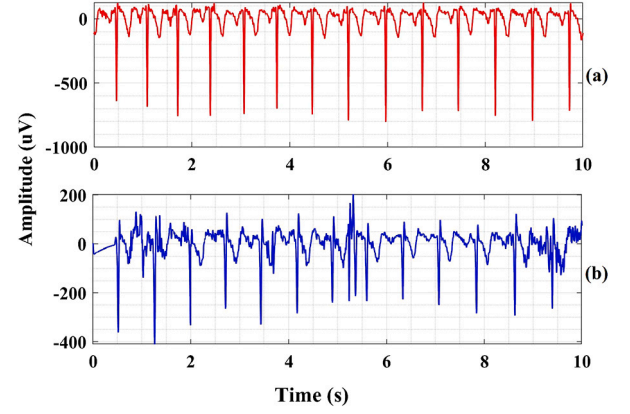


Fig. 2. ECG signals of two classes obtained from CAP dataset: (a) Normal (b) Insomnia.

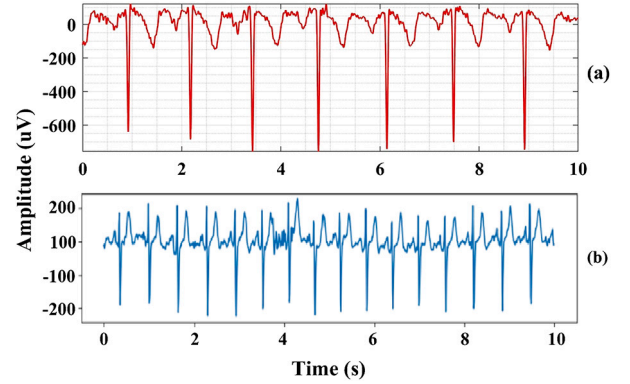


Fig. 3. ECG signals of two classes obtained from SDRG dataset: (a) Normal, and (b) Insomnia.

to analyze time-varying signals. It employs a variable length window to eliminate the fixed resolution problem of short-time Fourier transform (STFT) and decomposes a signal into a set of “wavelets” which can reveal T-F information about the input [26]. The CWT is performed by convolving the signal with a wavelet function that is scaled and translated in time. The result is a two-dimensional representation of a signal, where the x-axis represents time and the y-axis represents frequency [26]. This allows for the identification of patterns and features in the signal that may not be visible in the time or frequency domain alone. TFR extracted using CWT is called a scalogram. CWT of a signal $s(t)$ is computed as follows:

$$C_{\alpha,\beta}[s(t)] = \frac{1}{\sqrt{\alpha}} \int_{-\infty}^{\infty} s(t) \psi^* \left(\frac{t-\beta}{\alpha} \right) dt \quad (1)$$

where α indicates the scaling factor controls the function spread, β is a time shift and $\psi((t-\beta)/\alpha)$ denotes the shifted and scaled form of the mother wavelet $\psi(t)$. The CWT can be adapted to the signal characteristics by choosing the appropriate wavelet function, which allows for a better analysis of specific features of the input signal. This work utilizes the Morlet wavelet function as a mother wavelet to extract ECG scalograms using CWT. Figs. 4 and 5 are typical scalogram plots of two ECG classes (a: insomnia; b: normal) obtained from CAP and SDRG datasets, respectively.

2.3. Classification scheme

This work utilizes CNN architecture for the automated classification of ECG scalograms into normal and insomnia groups. The CNN model employs convolutional processes rather than matrix multiplication with the input image and evaluates deep features to distinguish one group

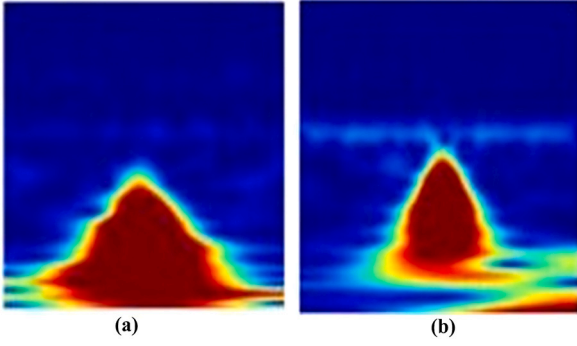


Fig. 4. Typical scalogram plots of two ECG classes obtained from CAP dataset: (a) Insomnia (b) Normal.

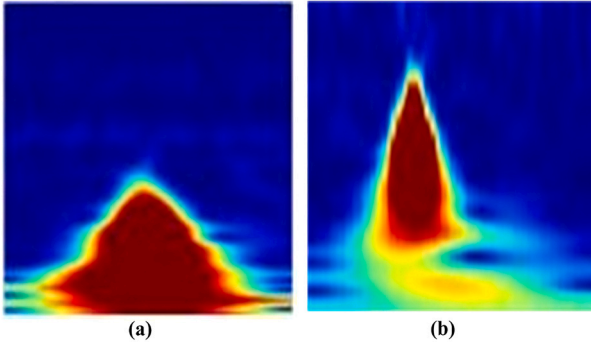


Fig. 5. Typical scalogram plots of two ECG classes obtained from SDRC dataset: (a) Insomnia, and (b) Normal.

from another. The CNN structure includes an input layer, numerous hidden layers, and a fully connected layer (FCL), with a softmax and classification layer. Each stage has a distinct function of analyzing the information acquired from the preceding layer. In the hidden layers, the features are detected and collected using several convolutions (CL) and pooling (PL) operations. To evaluate deep high-level characteristics, CL and PL are connected successively. The softmax function is used as the activation function in the output layer of neural network models that predict a multinomial probability distribution. At the end of a CNN, a classification layer is utilized to assign a class to an object [16]. Mathematically, a 2D-convolution operation is given as,

$$X \otimes Y(i, j) = \int A(x, y) B(x + i, y + j) \quad (2)$$

The output feature map dimension ($Fw_o \times Fh_o \times Fc_o$) using a $k \times k$ convolutional kernel, with padding of p , and stride of st is computed as,

$$\begin{aligned} Fw_o &= \frac{Iw_i + p - k}{st} + 1 \\ Fh_o &= \frac{Ih_i + 2p - k}{st} + 1 \end{aligned} \quad (3)$$

where, $Iw_i \times Ih_i \times c_i$ represents the size of an input image with c_i number of channels.

One can build their own CNN by combining the CL, PL, and FCL or can utilize the pre-trained deep CNNs. The number of CL, PL, and FCL in the model can be changed until the desired performance is attained. A conventional back-propagation approach is utilized to train the entire model. CNN testing and training require the fine adjustment of several hyperparameters. The CNN output can be adjusted by tuning several parameters, including weight, number, and size of kernels, learning and bias rate, etc.

In this study, three pre-trained models, AlexNet, MobileNetV2, and VGG16, along with a newly developed lightweight CNN for insomnia classification. AlexNet architecture was the first deep learning model designed to take advantage of the 2D structure of image data. It is com-

posed of 5 CLs and 3 FCLs [27]. The MobileNetV2 architecture consists of a series of inverted residual blocks, with each block containing a depthwise convolution followed by a pointwise convolution. The network also includes several linear bottlenecks and non-linear bottlenecks to reduce the number of parameters and increase the nonlinearity of the network. The final layers of MobileNetV2 include a global average PL with FCL for classification [28]. The VGG16 architecture consists of 16 layers, 13 CLs, and 3 FCLs. All the CLs use 3×3 filters with a stride of 1, and the PLs use 2×2 filters with a stride of 2. The architecture also makes use of very small convolutional filters and a large number of neurons in the FCLs [29]. Our proposed network consists of 8-CLs with ReLu activation function, 1-FCL, and 1-softmax layer along with a classification layer for the precise classification of normal and insomnia ECG scalograms. Fig. 6 shows the diagram of the developed neural network employed to classify ECG scalogram into insomnia and normal class.

3. Results

The main aim of this work is to develop a robust simple deep-learning model to detect insomnia using ECG signals. The CAP and SDRC sleep datasets have been employed to develop the proposed model. In this work, ECG data of 9-Insomnia patients and 16-normal subjects from CAP dataset; and 11-Insomnia patients and 11-normal subjects from SDRC dataset have been used to validate the proposed scheme. A total of 4,38,240 (181800 normal and 256440 insomnia) ECG epochs from the CAP dataset and 6,32,066 (320177 normal and 311889 insomnia) from the SDRC dataset have been utilized to build the models. The short window length gives better time resolution but poor frequency resolution. In contrast, wide-length windows provide high-frequency resolution but low-time resolution. The medium window provides improved resolution in both the temporal and frequency domains while avoiding overlapping. As a result, the Morlet wavelet function has been used as a mother wavelet to transform 1D ECG epochs into 2D scalograms using CWT [26]. Obtained scalograms are given to benchmark AlexNet, MobileNetV2, VGG16, and newly developed CNN. To match the dimension of the input all scalograms is resized to $227 \times 227 \times 3$ for AlexNet and $224 \times 224 \times 3$ for MobileNetV2, VGG16, and developed CNN. In this work, we first trained and validated the network on the CAP dataset and tested it on the SDRC dataset. Second time the model is trained and validated on the SDRC dataset and tested on the CAP dataset. We have used 85% of data for training, 15% of data for validation, and used another 100% of data for testing the model. Throughout the experiment, we maintained the same hyperparameters with setup. The Adam optimizer combines the greatest features of the AdaGrad and RMSProp algorithms while using less memory space [30]. Consequently, the Adam optimizer with a 0.001 initial learning rate is used, and 100 epochs are used with a batch size of 264. A total of 10 iterations/step is set per epoch. MaxPooling operation is performed in the pooling layers (PLs) and the Softmax function is used as an activation function. We have performed this experiment on a single graphics processing unit (GPU) Nvidia A700 with an i10 processor, 24-GB dedicated RAM, and 64-GB total central processing unit (CPU) RAM. Python3 and PyTorch deep learning frameworks have been utilized for software building. An overview of the various tuning parameters utilized by different CNN-based models is illustrated in Table 1. It can be understood from Table 1, that developed CNN requires 4.645 million learning parameters, which is less than AlexNet and VGG16 but more than MobileNetV2. The average time required to train each epoch of the presented CNN is only 7 minutes, which is less compared to all other networks. Thus, our developed model is lighter compared to AlexNet and VGG16 and faster than all currently utilized networks.

Tables 2 and 3 depicts the overall confusion matrix yielded for insomnia and normal scalogram classification of CAP and SDRC datasets using developed CNN, respectively.

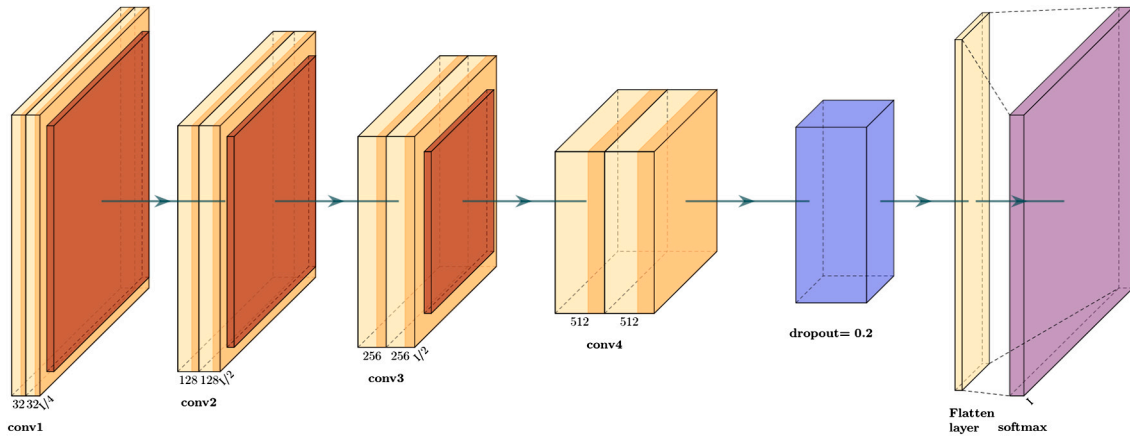


Fig. 6. Developed CNN model for automated insomnia detection.

Table 1

Overview of the tuning parameters used for various CNN-based models.

Parameters	AlexNet	VGG16	MobileNetV2	Proposed CNN
CLs	5	16	27	8
Filter size	11,5,3	3	3,1	3
Strides	1,4	1	2,1	1
FCLs	3	2	1	1
TLPs (in million)	61	138	3.47	4.645
Avg. TREE (in minute)	13	45	17	7

TLPs: Total learning parameters.

Avg. TREE: Average time required to train each epoch.

Table 2

Confusion matrix yielded for CAP dataset using developed CNN.

Class	Insomnia	Normal
Insomnia	98.50%	1.50%
Normal	0.80%	99.20%

Table 3

Confusion matrix yielded for SDRC dataset using developed CNN.

Class	Insomnia	Normal
Insomnia	99.34%	0.66%
Normal	2%	98%

The effectiveness of the employed networks is evaluated using six performance measures: accuracy (ACC), false omission rate (FOR), sensitivity (SEN), false discovery rate (FDR), specificity (SPE), and critical success index (CSI) / threat score (TS). These performance parameters are mathematically described as follows:

$$ACC(\%) = \frac{T_P + T_N}{T_P + F_N + T_N + F_P} \times 100 \quad (4)$$

$$FOR = \frac{F_N}{T_N + F_N} \quad (5)$$

$$SEN(\%) = \frac{T_P}{T_P + F_N} \times 100 \quad (6)$$

$$FDR = \frac{F_P}{T_P + F_P} \quad (7)$$

$$SPE(\%) = \frac{T_N}{T_N + F_P} \times 100 \quad (8)$$

$$CSI = \frac{T_P}{T_P + F_N + F_P} \quad (9)$$

where T_P and T_N represent correctly detected positive and negative occurrences, respectively. Similarly, the incorrectly detected number

Table 4

Performance measures (%) obtained for various CNN-based models using CAP dataset.

Parameters	AlexNet	MobileNetV2	VGG16	Proposed CNN
ACC	97.35	96.59	96.82	98.91
FOR	2.79	3.70	3.30	1.50
SEN	98.01	97.36	97.64	98.94
FDR	2.55	3.20	3.10	0.80
SEP	96.43	95.52	95.67	98.87
CSI	0.96	0.94	0.95	0.98

Table 5

Performance measures (%) obtained for various CNN-based models using SDRC dataset.

Parameters	AlexNet	MobileNetV2	VGG16	Proposed CNN
ACC	97.98	96.06	96.67	98.68
FOR	1.06	3.65	3.22	0.66
SEN	98.89	96.23	96.69	99.31
FDR	3.00	4.23	3.45	2.00
SEP	97.13	95.90	96.64	98.08
CSI	0.96	0.92	0.93	0.97

of positive and negative episodes is represented by F_P and F_N , respectively. Table 4 represents the performance measures obtained for various CNN-based models using CAP dataset.

Table 5 shows the overall performance measures obtained for various CNN-based models using the SDRC dataset. Tables 4 and 5 justify the superiority of our proposed model in the automated detection of insomnia using both CAP and SDRC sleep datasets. Figs. 7 and 8, represent the bar plots showing the performance matrices (%) obtained for various deep learning models using CAP and SDRC data, respectively.

It may be noted from Tables 4, 5, and Figs. 7 and 8 that the proposed CNN attained higher performance metrics compared to other pre-trained models.

To assess the robustness of our developed system, we have computed the standard deviation (std) at a 95% confidence interval of the performance metrics obtained from all the deep learning models for both the CAP and SDRC datasets. Table 6 presents the std values at a 95% confidence interval for various performance parameters. Hence, the proposed network is more reliable than AlexNet, MobileNetV2, and VGG16.

Fig. 9 (a) and (b) depict the accuracy v/s epochs and loss v/s epochs graphs for the developed model using CAP and SDRC datasets, respectively. The developed model performs well on both the CAP and SDRC datasets, as shown in the aforementioned Fig. 9 (a) and (b), respec-

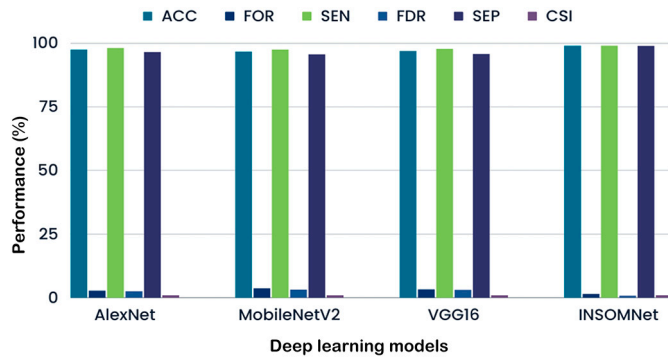


Fig. 7. Bar plots showing the performance matrices (%) obtained for various deep learning models using CAP dataset.

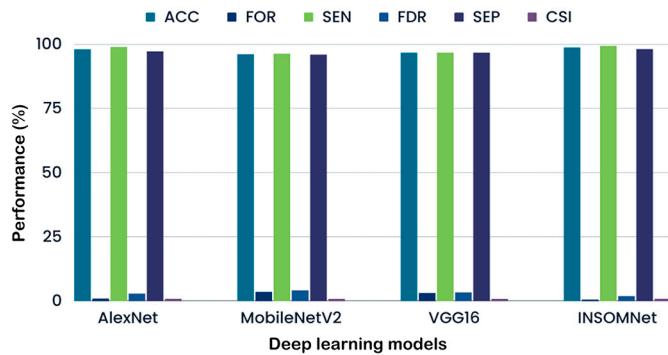


Fig. 8. Bar plots showing the performance matrices (%) obtained for various deep learning models using SDRC dataset.

Table 6

Standard deviation values (at a 95% confidence interval) obtained by different deep learning models for various performance parameters.

Parameters	AlexNet	MobileNetV2	VGG16	Proposed CNN
ACC	0.633	0.529	0.150	0.231
FOR	1.730	0.050	0.080	0.840
SEN	0.880	1.127	0.953	0.374
FDR	0.450	1.030	0.350	1.200
SEP	0.700	0.377	0.970	0.791
CSI	0.004	0.020	0.012	0.008

tively. However, the model performed better and diverged less for the CAP dataset compared to the SDRC dataset (smaller difference between training and validation accuracy curves).

4. Discussion

The above results represent that the ECG scalograms extracted using CWT contain more appropriate details about the signals, and are suitable to detect insomnia patients by employing a CNN architecture. The handcrafted feature extraction and classification have been greatly reduced due to automated deep feature detection and classification by CNN architecture. Using a minimum number of layers and fewer tuning parameters, the presented network with ECG scalogram has attained higher classification accuracy compared to the pre-trained deep learning models (AlexNet, MobileNetV2, VGG16).

There are some physiological and cerebral activations between patients with insomnia and normal subjects [31]. Work done using power spectra-based characteristics has indicated many outcomes between insomniac and normal classes [32]. Several studies have indicated that a person with insomnia has a higher heart rhythm and blood pressure than healthy subjects [33,34]. Periodic limb movement disorder has

Table 7

Summary of performance comparison with state-of-the-art techniques developed for automated detection of insomnia using ECG signals.

Authors	Dataset used	Presented method	Performance (in %)
Sharma et al. [2]	CAP sleep	ECG features and KNN classifier	ACC = 97.87
Tripathi et al. [36]	CAP sleep	PSD from HRV and RF classifier	ACC = 96
Urtnasan et al. [37]	CAP sleep	Sleep disorder network	F-1 score = 95
Proposed system	CAP and SDRC sleep	ECG scalogram and CNN	ACC = 98.91 on CAP ACC = 98.68 on SDRC

been detected using ECG signals and deep learning techniques [35]. Actigraphy time series data of multiple nights have been utilized to detect acute insomnia [18]. In [2], authors depicted that the values of $L - 1$ norm for insomnia patients are lower than normal subjects.

A summary of performance comparisons with state-of-the-art techniques developed for automated detection of insomnia using ECG signals is given in Table 7. Sharma et al. [2] decomposed the ECG signals using an optimal antisymmetric BW filter bank and computed $L - 1$ norms features from each subband. They attained an ACC of 97.87% in detecting insomnia using KNN classifier. Tripathi et al. [36] extracted the heart rate variability (HRV) signals from the ECG data and evaluated the power spectral density (PSD). They applied PSD features to the random forest (RF) classifier and yielded an ACC of 96%. Urtnasan et al. [37] developed a sleep disorder network for the classification of different sleep stages using ECG signals. Their developed network attained an F-1 score of 95% for the detection of periodic leg movement. In previous studies, machine learning techniques have been employed for automated sleep disorder detection. These approaches typically involve extracting a range of features from the ECG data, including nonlinear, time-domain, and frequency-domain features, to facilitate classification. However, the process of selecting the most appropriate features can be time-consuming and manual. Additionally, machine learning-based methodologies often require careful tuning of parameters for signal decomposition and feature extraction. These tasks further add to the complexity and challenges associated with developing accurate and robust insomnia detection models using machine-learning approaches. Our present study proposed an intelligent system using an integration of an ECG scalogram and a CNN approach for the precise detection of insomnia patients. Our developed INSOMNet obtained the highest classification ACC of 98.91% using the CAP dataset and 98.68% using the SDRC dataset. This justifies that the scalogram coupled with CNN, can pick the subtle variations present in the ECG signals and yield the highest performance. Also, the acquisition of ECG signals is simpler than EEG signals. Also, our proposed method can be used in home-based detection and also can be integrated into wearable devices to detect insomnia automatically.

The aforementioned discussion makes it evident that the proposed system in combination with ECG scalograms performed better than the previous state-of-art techniques and is best suitable for the accurate detection of insomnia patients. Our developed CNN is less complex, faster, and performed better than AlexNet, MobileNetV2, and VGG16 models. The presented system yielded the highest classification ACC of 98.91% and 98.68% on CAP and SDRC sleep datasets, respectively. To the best of our knowledge, this is the first study to diagnose insomnia using two sleep datasets (CAP and SDRC) and a deep learning model. Our results indicate that ECG signals can be used to detect insomnia instead of EEG signals. These ECG signals have information about the brain. Brain and heart interactions are mediated by the autonomous nervous

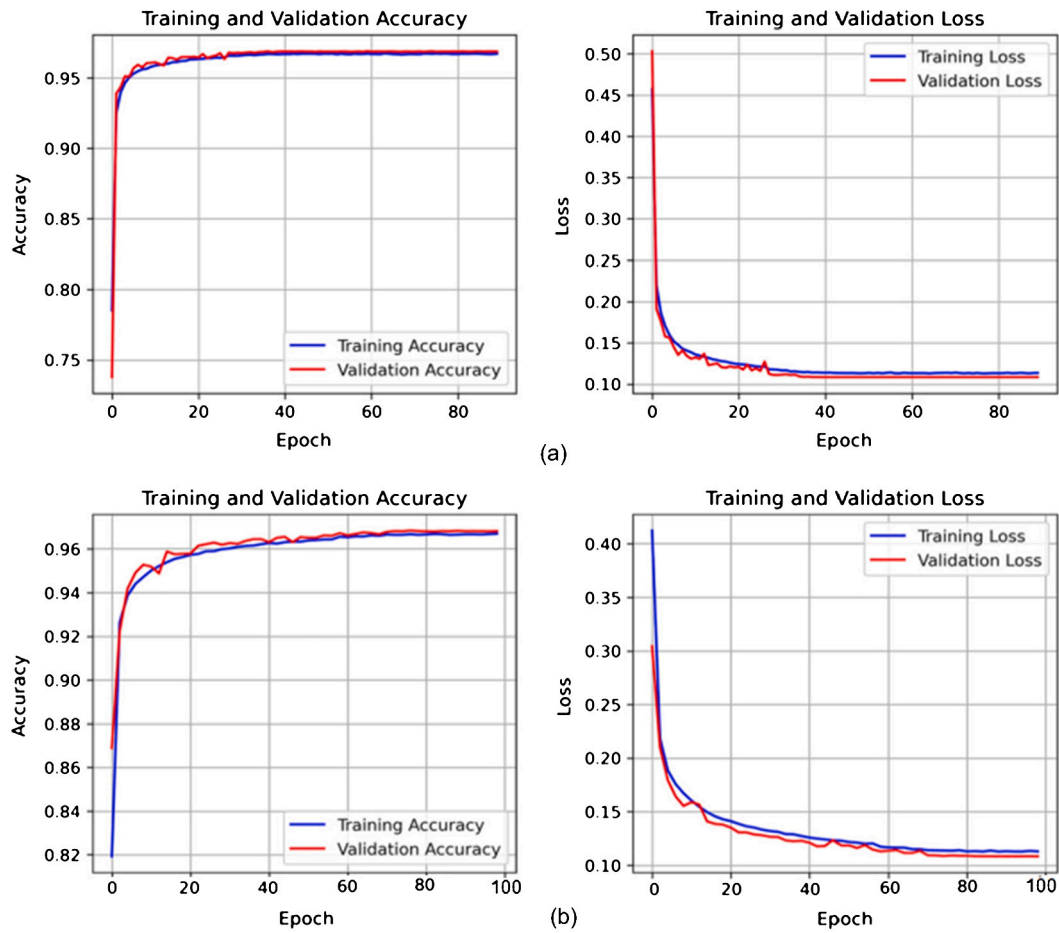


Fig. 9. Graphical representation of accuracy v/s epochs and loss v/s epochs obtained for the proposed model of two datasets: (a) CAP, and (b) SDRC.

system in healthy and diseased patients [38–40]. Hence, widely available, less noisy, and inexpensive ECG signals can be employed instead of EEG signals for insomnia detection.

The key benefits of the presented automated insomnia detection system are as follows:

- The developed INSOMNet is robust and reliable since it is trained and tested on two datasets of different sampling frequencies.
- Denoising of input ECG signals is not required.
- The presented system has the potential to detect other types of sleep disorders with the highest classification performance.
- The approach is straightforward, less complicated, quick, and fully automated as separate feature extraction and classification are not required.
- As ECG signals have been used to identify insomnia, the proposed model seems a suitable candidate for a wearable device to measure the quality of sleep.

The limitations of the proposed system are given below:

- Utilized the empirical parameters.
- Employed few participants in normal and insomnia groups of our proposed study.
- Computationally intensive as we need to convert the ECG signal into scalograms.

Our developed ECG-based model needs to be tested with huge diverse datasets collected from various centers before employing for clinical usage. This is the first deep learning-based study to be employed in automated insomnia detection using ECG signals. To obtain the trust of the

clinicians we plan to incorporate the explainable artificial intelligence (XAI) into our proposed model [41,42]. This will help to provide more information like heat maps for the clinicians and justify our diagnosis. Also, we intend to employ uncertainty quantification in our developed model using ECG signals. This will help to develop a trustworthy model in clinical scenarios.

5. Conclusion

In this study, we presented a novel INSOMNet for the automated identification of insomnia using the ECG scalogram coupled with a CNN model. The proposed system developed using ECG sleep signals outperformed the existing EEG, EoG, and PSG-based methods. The obtained scalograms from CWT coupled with CNN yielded higher performance, on publicly available CAP and SDRC datasets. The developed system attained the classification ACC of 98.91% and 98.68% for CAP and SDRC datasets, respectively. CWT can extract more minute information from input ECG signals. It collects both temporal and spatial information simultaneously, which makes it extremely useful in discriminating insomnia from normal class using ECG signals. Our developed network is lightweight, less complex, and more accurate compared to the pre-trained models. The proposed system is less complex and fully automated. The prototype is suitable to design a wearable for screening insomnia that can increase the comfort level in daily life. The proposed model can help sleep experts in measuring the quality of sleep. In the future, we plan to extend our developed model to detect other sleep-related disorders such as periodic leg movement, sleep apnea, narcolepsy, rapid-eye movement, bruxism, and narcolepsy.

Declaration of competing interest

None.

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